

Roll No.

TPE-004

**B. TECH. (SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022
DIGITAL SYSTEM USING VERILOG**

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each question carries 10 marks.

1. (a) What are System Tasks in Verilog ? Explain with at least *two* examples.
(CO1)

OR

- (b) Explain the following data types with examples : Nets, Register, Vector, String and Time.
(CO1)
2. (a) Discuss the different levels of abstraction used in Verilog modelling using suitable examples.
(CO1)

OR

- (b) Explain the port connection rules of Verilog HDL with examples. Draw the block diagram mentioning components of a Verilog module.
(CO1)

3. (a) $A = 5'b10101$, $B = 5'b11111$, $C = 5'b11000$, $D = 5'b10001$.
Evaluate :
(CO2)

(i) $A \& B$

(ii) $C \wedge D$

P. T. O.

(2)

(iii) $B \% D$

(iv) $B << 2$

(v) $!(&A)$

(vi) $Y = \{A \geq B\}$

(vii) $Y = \{A[2], B[0], C\}$

(viii) $Y = (A > B) : 1 : 0$

OR

(b) Using Gate-Level modelling write the Verilog code for 4-bit adder/Subtractor circuit. (CO2)

4. (a) What is continuous assignment ? Write a Verilog code for 4×1 MUX using ternary operators. (CO2)

OR

(b) Differentiate with the help of examples : (CO2)

(i) always and initial procedural statements

(ii) blocking and non-blocking statements

5. (a) Write a Verilog code to realize 2-bit comparator (A_1A_0 & B_1B_0) in gate level to give the outputs $A_{\text{equal}} B$, $A_{\text{greater than}} B$, $A_{\text{less than}} B$. Verify the code with an appropriate test bench. (CO2)

OR

(b) Write a Verilog code to generate the following sequence 1, 2, 4, 6, 7, 1, 2 (CO2)